

BATTULA GURUPRASAD SPOORTHY

B.Tech in Computer Science and Engineering
Vellore Institute of Technology, Amaravati

bgpspoorthi@gmail.com — +91 7601053542 — LinkedIn — GitHub

Career Profile

Computer Science undergraduate specializing in Machine Learning and Deep Learning with hands-on experience in building real-time AI systems and transformer-based applications. Skilled in computer vision, semantic similarity modeling, and model optimization with a strong focus on solving practical business problems using data-driven approaches. Experienced in Generative AI techniques and cloud-based deployment, with the ability to translate complex problem statements into scalable AI solutions.

Education

B.Tech in Computer Science and Engineering, VIT-AP University	2023 – 2027
Senior Secondary, Sri Chaitanya Junior College	2021 – 2023
Secondary School, Taksha Shila High School	2021

Technical Skills

Programming: C, C++, Java, Python, SQL, JavaScript
Core CS: Data Structures, Algorithms, OOP, SDLC
Web & Frontend: HTML, CSS
Databases: MySQL, Relational Database Management Systems (RDBMS)
Libraries & Frameworks: NumPy, Pandas, OpenCV, TensorFlow, Keras
Cloud & Tools: AWS (Fundamentals), VS Code, Jupyter Notebook, Google Colab

Non-Technical Skills

Core Strengths: Analytical Thinking, Problem-Solving, Logical Reasoning
Communication: Effective Verbal and Written Communication
Collaboration: Teamwork, Cross-functional Coordination
Personal Traits: Adaptability, Leadership, Time Management

Experience

- SDE Intern, BlueStock Fintech** June – July 2025
- Developed and deployed an IPO management portal with modular components for application tracking and investment management.
 - Improved system responsiveness and optimized UI performance across desktop and mobile platforms.
- Co-Lead, Blockchain Club** 2024 – Present
- Led technical initiatives and organized workshops on blockchain fundamentals and decentralized technologies.
 - Designed problem statements and coordinated technical events to promote hands-on blockchain development.

Research Experience

- Deep Learning-Based Real-Time Emotion Analytics for Multi-Person Settings**
- Designed a two-stage FER pipeline using MTCNN for multi-face detection and lightweight CNN for emotion classification.
 - Integrated confidence score visualization using softmax probabilities.
 - Achieved real-time performance (28 FPS) with optimized inference pipeline suitable for live multi-person retail and surveillance environments.
 - Tech Stack:** Python, TensorFlow/Keras, OpenCV, CNNs, MTCNN

Projects

AI-Powered Smart Duplicate File Detection System *Python, Streamlit*

- Developed an intelligent duplicate detection system using SHA-256 hashing for exact matches and perceptual image hashing for visual similarity detection.
- Integrated transformer-based semantic similarity models for advanced text comparison and near-duplicate detection.
- Designed similarity threshold tuning and embedding-based semantic comparison to reduce false positives in near-duplicate detection.
- Designed an interactive web interface with storage analytics and secure file management features.

Object Counter

C++, Arduino, Computer Vision

- Built a real-time object detection and counting system using computer vision techniques integrated with Arduino-based hardware.
- Enabled automated counting with hardware-triggered responses for smart monitoring applications.

Food Banking Website

Full Stack Web Application

- Developed a full-stack web platform connecting food donors with recipients to streamline food redistribution.
- Implemented logic for request handling, authentication, and efficient database management.

WaveHand – Gesture-Based Smart Home Automation

Python, OpenCV, C++, Arduino

- Implemented real-time hand gesture recognition using OpenCV for contactless appliance control.
- Integrated Arduino Uno with relay modules to automate home devices based on gesture input.

Achievements

- Team Member, VITOPIA — Assisted in organizing and executing large-scale campus fest events, supporting coordination and smooth event operations.
- Selected for Round 2 in a competitive Hackathon for innovative team-based problem solving.

Certifications

- AWS Academy Graduate – Cloud Architecting Certificate
- AWS Certified Cloud Practitioner
- Introduction to Generative AI Studio — Google Cloud
- MATLAB Certificate